Gen AI Engineer

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PROFESSIONAL SUMMARY:

- **Senior Generative AI & ML Engineer** with 10+ years of progressive experience, **advancing from Data Engineering through ML Engineering to Senior AI Engineer**, delivering **enterprise-scale AI systems** across **finance, retail, insurance, and healthcare**.
- **Architected enterprise-scale Generative AI platforms** leveraging **LLMs** (**GPT-4, Claude, LLaMA**), **RAG architectures**, **LangChain, LangGraph**, **Vector Databases** (**Pinecone, FAISS, Weaviate**), **embeddings**, and **prompt engineering**; deployed solutions via **AWS Bedrock, SageMaker, Lambda, ECS**.
- **Built scalable Machine Learning pipelines** with **TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, MLflow, Airflow**, and productionized models using **Docker, Kubernetes (EKS)**, **CI/CD pipelines** (**Jenkins, GitHub Actions, GitLab CI**), ensuring **high-performance AI deployment**.
- **Engineered high-volume Data Engineering frameworks** with **Apache Spark, PySpark, Kafka, Hadoop, Hive, Snowflake, Redshift, Databricks**, implementing robust **ETL/ELT pipelines** for **structured and unstructured data**.
- **Developed full-stack AI applications** using **Python, FastAPI, Flask, Node.js, ReactJS, Angular, TypeScript**, **REST/GraphQL APIs**, **OAuth2, JWT, SSO**, and **microservices architecture** within **secure enterprise environments**.
- **Led MLOps/LLMOps initiatives**, including **model monitoring, drift detection, Prometheus, Grafana, CloudWatch, automated retraining, feature stores**, and **cost optimization via FinOps**, while ensuring **real-time, low-latency AI inference**.
- **Built advanced NLP and Semantic AI systems** using **Transformers, BERT, Hugging Face, OpenAI APIs**, **fine-tuning, semantic search**, and **hybrid search architectures** for **financial and retail analytics**.
- **Integrated AI solutions with legacy systems**, including **mainframes, COBOL services, Oracle DB, SQL Server, Informatica, and batch frameworks**, ensuring **seamless hybrid architecture interoperability**.
- **Developed enterprise backend services** with **Java, Spring Boot, API Gateway, Load Balancers, Auto Scaling Groups**, and implemented secure **SOAP/REST integrations**; automated infrastructure via **Terraform, CloudFormation, IaC**, managing **EC2, S3, EKS, RDS**, and **multi-account cloud governance**.
- **Implemented secure AI infrastructure** using **IAM, RBAC, VPC, Secrets Manager, Data Masking, PII compliance**, and **Model Governance frameworks** aligned with **regulatory standards**.
- **Designed enterprise data warehousing and BI solutions** with **dimensional modeling, star/snowflake schemas, SQL optimization, data quality frameworks, metadata management**, reporting via **Tableau and Power BI**.
- **Collaborated in Agile/Scrum environments**, conducted **code reviews, version control via Git/Bitbucket**, enforced **unit testing (PyTest, JUnit)**, developed **automation/monitoring scripts** (**Python, Bash, PowerShell, ELK Stack**), ensuring **reliable multi-region production deployments**.

TECHNICAL SKILLS:

Category	Skills & Tools
Generative AI	GPT-4, Claude, Llama, RAG, LangChain, LangGraph, Prompt Engineering, Embeddings, Vector Databases (Pinecone, FAISS, Weaviate), OpenAI APIs
Machine Learning	TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, MLflow, Model Monitoring, Drift Detection, Feature Engineering
Data Engineering	Apache Spark, PySpark, Kafka, Hadoop, Hive, Databricks, Snowflake, Amazon Redshift, ETL/ELT Pipelines
Backend Development	Python, FastAPI, Flask, Java, Spring Boot, Node.js, REST APIs, GraphQL, Microservices Architecture
Frontend Development	Angular, ReactJS, TypeScript, HTML5, CSS3
Cloud Platforms	AWS (Bedrock, SageMaker, Lambda, ECS, EKS, EC2, S3, RDS, IAM, VPC, KMS), Azure, GCP, Multi-Account Governance
DevOps & CI/CD	Docker, Kubernetes, Terraform, CloudFormation, Jenkins, GitHub Actions, GitLab CI
Databases	PostgreSQL, MySQL, Oracle, SQL Server, MongoDB, DynamoDB, Redis
Streaming & Real-Time	Apache Kafka, AWS Kinesis, Event-Driven Architecture
Security & Compliance	IAM, RBAC, OAuth2, JWT, SSO, Data Masking, PII Compliance, Encryption
Monitoring & Observability	Prometheus, Grafana, CloudWatch, ELK Stack
BI & Reporting	Power BI, Tableau
Version Control & Agile	Git, Bitbucket, Agile, Scrum

PNC Financial Services – New York, NY **May 2024 – Present**
Gen AI Engineer
AI-Driven Customer Insights & Fraud Detection Platform

Responsibilities:

- Architected **enterprise Generative AI platforms** using **AWS** (Bedrock, SageMaker, EKS, EC2, S3, RDS, Lambda, IAM, VPC, CloudWatch) for **scalable AI deployments** in fraud detection and customer insights.
- Designed **advanced retrieval pipelines** integrating **vector search, hybrid retrieval, and contextual re-ranking** for financial data analysis.
- Leveraged **LLMs and transformer-based models** for **domain-aware AI query processing** across financial and behavioral datasets.
- Built **embedding pipelines** using **BERT-based models** with **Pinecone and FAISS** for semantic search.
- Developed **unified retrieval systems** using **Snowflake, Snowflake Cortex, and PostgreSQL** for structured and unstructured enterprise data.
- Engineered **scalable data pipelines** with **Apache Spark, Kafka, and Airflow** for batch and real-time AI processing.
- Built **high-throughput Python/FastAPI microservices** with async APIs, increasing **inference throughput by 40%** and reducing latency by 32% across fraud detection workloads.
- Developed **Python automation scripts** for **data processing, ETL, monitoring, and workflow orchestration**.
- Designed **secure authentication** using **OAuth2 and JWT**, ensuring enterprise-grade API security and compliance.
- Built **interactive Angular dashboards**, improving **fraud alert visibility by 50%** and reducing analyst investigation time by 30%.
- Implemented **model evaluation and optimization** using **RAGAS**, improving **faithfulness, relevancy, and reducing hallucinations**.
- Built **ML/DL pipelines** using **TensorFlow and PyTorch**, adapting models for fraud detection and customer behavior analytics.
- Managed **model lifecycle and experimentation** using **MLflow**, including drift detection, retraining workflows, and performance tracking.
- Containerized services with **Docker** and deployed via **Kubernetes (EKS)** for scalable microservices architecture.
- Automated **infrastructure provisioning** using **Terraform (IaC)** and implemented **multi-account cloud governance**.
- Designed and maintained **CI/CD pipelines** for automated testing, secure builds, and production deployments.
- Implemented **Redis caching** to reduce latency for **real-time AI inference and fraud detection workflows**.
- Established **observability frameworks** using **Prometheus, Grafana, and CloudWatch** to monitor AI performance and system health.
- Delivered **full lifecycle AI solutions**, covering **data ingestion, retrieval, model inference, evaluation, deployment, and monitoring**.
- Collaborated in **Agile/Scrum environments**, conducted **code reviews, enforced unit testing (PyTest, JUnit)**, and ensured **reliable production deployments across multi-region cloud systems**.

Environment: AWS (SageMaker, EKS, EC2, S3, RDS, Lambda, IAM, VPC, CloudWatch), Snowflake, Snowflake Cortex, PostgreSQL, Python, FastAPI, Angular, BERT, FAISS, Apache Spark, Kafka, Airflow, TensorFlow, PyTorch, MLflow, Docker, Kubernetes, Terraform, CI/CD, Prometheus, Grafana, OAuth2, JWT, Redis

Humana – Louisville, KY **Sept 2022 – April 2024**
Senior AI/ML Engineer
Patient Health Data Analytics & Predictive Care System
Responsibilities:

- Architected **enterprise AI/ML platforms** using **Azure** (Machine Learning, AKS, Virtual Machines, Blob Storage, SQL Database, Functions, Key Vault, Virtual Network, Monitor) for **predictive patient care and health data analytics**.
- Designed **advanced retrieval pipelines and hybrid search architectures** for **intelligent clinical data insights**.
- Leveraged **LLMs and transformer-based models** for **domain-aware patient health queries and predictive analytics**.
- Built **embedding pipelines** using **BERT-based models** with **Pinecone and FAISS** for semantic search across patient records and clinical notes.
- Developed **unified retrieval systems** using **Snowflake and PostgreSQL** for structured and unstructured healthcare datasets.
- Engineered **scalable data pipelines** using **Apache Spark, Kafka, and Airflow** for batch and real-time patient analytics.
- Built **backend microservices** using **Python and FastAPI** for **high-throughput AI inference and predictive care automation**.
- Developed **interactive dashboards** using **ReactJS**, delivering **predictive health alerts, patient risk scoring, and operational insights**.
- Designed **secure authentication** using **OAuth2 and JWT**, ensuring **HIPAA-compliant API access and data security**.
- Implemented **model evaluation and optimization** using **custom metrics**, improving **faithfulness, relevancy, and accuracy** of patient risk predictions.
- Built **ML/DL pipelines** using **TensorFlow and PyTorch**, adapting models for **patient outcome prediction and preventive care**.
- Managed **model lifecycle and experimentation** using **MLflow**, including drift detection, retraining workflows, and performance monitoring.
- Containerized services with **Docker** and deployed via **Azure Kubernetes Service (AKS)** for **scalable, resilient microservices architecture**.
- Automated **infrastructure provisioning** using **Terraform (IaC)** and implemented **multi-account cloud governance**.
- Designed and maintained **CI/CD pipelines**, enabling automated testing, secure builds, and production deployments.
- Implemented **Redis caching** to reduce latency for **real-time predictive care alerts and AI inference**.
- Established **observability frameworks** using **Azure Monitor, Prometheus, and Grafana** for monitoring **AI model performance and system health**.

- Delivered **end-to-end AI/ML solutions**, covering **data ingestion, retrieval, inference, evaluation, deployment, and monitoring**, while integrating **ReactJS dashboards** for **real-time clinical and operational insights**.

Environment: Azure (Machine Learning, AKS, Virtual Machines, Blob Storage, SQL Database, Functions, Key Vault, Virtual Network, Monitor), Snowflake, PostgreSQL, ReactJS, FastAPI, Python, BERT, FAISS, Apache Spark, Kafka, Airflow, TensorFlow, PyTorch, MLflow, Docker, Kubernetes, Terraform, CI/CD, Prometheus, Grafana, OAuth2, JWT, Redis, HIPAA Compliance

Target Corporation – Minneapolis, MN **April 2019 – August 2022**
AI/ML Engineer
AI Demand Forecasting & Personalization Platform

Responsibilities:
- Architected **enterprise AI/ML platforms** using **GCP** (Vertex AI, GKE, Compute Engine, Cloud Storage, Cloud SQL, Cloud Functions, IAM, VPC, Cloud Monitoring) for **demand forecasting and personalized shopping experiences**.
- Designed **advanced retrieval pipelines and hybrid search architectures** for **retail analytics**.
- Leveraged **ML and transformer-based models** for **product recommendation, demand prediction, and customer personalization**.
- Built **embedding pipelines** using **BERT-based models** with **Pinecone and FAISS** for semantic search of product catalogs and user behavior data.
- Developed **unified retrieval systems** using **Snowflake, Snowflake Cortex, and PostgreSQL** for structured and unstructured retail datasets.
- Engineered **scalable data pipelines** using **Apache Spark, Kafka, and Airflow** for batch and real-time AI processing.
- Built **backend microservices** using **Python and FastAPI** for high-throughput inference on purchase patterns and personalization models.
- Developed **interactive dashboards** using **Angular** and **Power BI**, delivering real-time **demand forecasts, inventory alerts, and personalization insights**.
- Designed **secure authentication** using **OAuth2 and JWT**, ensuring **enterprise-grade API security** and compliance.
- Implemented **model evaluation and optimization** using **custom metrics and pipeline validation** for **recommendation relevance, accuracy, and quality**.
- Built **ML/DL pipelines** using **TensorFlow and PyTorch** for **sales prediction, product affinity, and inventory optimization**.
- Managed **model lifecycle and experimentation** using **MLflow**, including drift detection, retraining workflows, and performance monitoring.
- Containerized services with **Docker** and deployed via **GKE**, automating **infrastructure provisioning** using **Terraform (IaC)**.
- Designed and maintained **CI/CD pipelines**, enabling automated testing, secure builds, and production deployments.
- Implemented **Redis caching** and established **observability frameworks** using **Prometheus, Grafana, and Cloud Monitoring** for real-time AI performance monitoring.
- Delivered **end-to-end AI solutions**, covering **data ingestion, retrieval, model inference, evaluation, deployment, and monitoring**, with **Python backend** and **Angular dashboards** for actionable retail insights.

Environment:GCP (Vertex AI, GKE, Compute Engine, Cloud Storage, Cloud SQL, Cloud Functions, IAM, VPC, Cloud Monitoring), Snowflake, PostgreSQL, Python, FastAPI, Angular, Power BI, BERT, FAISS, Apache Spark, Kafka, Airflow, TensorFlow, PyTorch, MLflow, Docker, Kubernetes, Terraform, CI/CD, Prometheus, Grafana, OAuth2, JWT, Redis

Travelers Companies – Hartford, CT **September 2017 – March 2019**
Senior Data Science Engineer
Insurance Claim Prediction & Risk Analytics Platform

Responsibilities:
- Architected **enterprise AI/ML platforms** using **AWS** (SageMaker, EC2, S3, RDS, Lambda, IAM, VPC, CloudWatch) for **insurance claim prediction and risk analytics**.
- Designed **advanced retrieval pipelines** for claims and policy data insights.
- Leveraged **ML and deep learning models** for **policy analysis, fraud detection, and risk assessment**.
- Built **embedding pipelines** using **BERT-based models** with **Pinecone and FAISS** for semantic search across claims and policy documents.
- Developed **unified retrieval systems** using **Snowflake, Snowflake Cortex, and PostgreSQL** for structured and unstructured insurance data.
- Engineered scalable **data pipelines** using **Apache Spark, Kafka, and Airflow** for batch and real-time insurance analytics.
- Built **backend microservices** using **Python and FastAPI** for **high-throughput predictive inference**.
- Developed **interactive dashboards** using **ReactJS**, delivering **real-time risk scoring, fraud alerts, and claims insights**.
- Designed **secure authentication** using **OAuth2 and JWT**, ensuring **enterprise-grade API security** and compliance.
- Built **ML/DL pipelines** using **TensorFlow and PyTorch** for **claims prediction, risk scoring, and fraud detection**.

- Managed **model lifecycle and experimentation**, including **drift detection, retraining workflows, and performance monitoring**.
- Containerized services with **Docker** and deployed via **Kubernetes (EKS)** for **scalable microservices architecture**.
- Automated **infrastructure provisioning** using **Terraform (IaC)** and maintained **CI/CD pipelines** for testing and deployments.
- Implemented **Redis caching** and established **observability frameworks** using **Prometheus, Grafana, and CloudWatch** to monitor **AI model performance and system health**.

Environment: AWS (SageMaker, EKS, EC2, S3, RDS, Lambda, IAM, VPC, CloudWatch), Snowflake, PostgreSQL, Python, FastAPI, ReactJS, BERT, FAISS, Apache Spark, Kafka, Airflow, TensorFlow, PyTorch, MLflow, Docker, Kubernetes, Terraform, CI/CD, Prometheus, Grafana, OAuth2, JWT, Redis

Krios Info Solutions – Pune, India **June 2015 – April 2017**
Data Engineer
Healthcare Claims Processing & Analytics Platform

Responsibilities:
- Architected scalable **AI/ML data pipelines** using **Azure** (Virtual Machines, Azure ML, Blob Storage, SQL Database, Functions, Key Vault, Virtual Network, Monitor) for **healthcare claims processing and analytics**.
- Designed **hybrid search and retrieval pipelines** for claims data insights.
- Leveraged **ML and deep learning models** for **claims validation, anomaly detection, and predictive analytics**.
- Built **embedding and semantic search pipelines** using **BERT-based models** for structured and unstructured claims data.
- Developed **unified retrieval systems** using **Snowflake and PostgreSQL** for efficient healthcare data management.
- Engineered **ETL pipelines** using **Apache Spark, Kafka, and Airflow** for batch and real-time claims processing.
- Built **backend microservices** using **Python and FastAPI** for **high-throughput AI inference and claims automation**.
- Developed **interactive dashboards** using **ReactJS**, delivering **real-time claims analytics, KPIs, and anomaly alerts**.
- Designed **secure authentication** using **OAuth2 and JWT**, ensuring **HIPAA-compliant access and data security**.
- Built **ML/DL pipelines** using **TensorFlow and PyTorch** for **claims scoring, risk assessment, and predictive analytics**.
- Managed **model lifecycle and experimentation** using workflow automation for **drift detection, retraining, and monitoring**.
- Containerized services with **Docker** and deployed on **Azure cloud** for **scalable microservices architecture**.
- Automated infrastructure provisioning using **IaC best practices** and maintained **CI/CD pipelines** for testing and deployments.
- Established **observability frameworks** using **Azure Monitor, Prometheus, and Grafana** to monitor **system health and AI model performance**.

Environment: Azure (Virtual Machines, Azure ML, Blob Storage, SQL Database, Functions, Key Vault, Virtual Network, Monitor), Snowflake, PostgreSQL, Python, FastAPI, ReactJS, BERT, FAISS, Apache Spark, Kafka, Airflow, TensorFlow, PyTorch, Docker, CI/CD, Prometheus, Grafana, OAuth2, JWT, Redis, HIPAA Compliance

EDUCATION:
Bachelor of Technology in Computer Science
Kakatiya College, Telangana 2011-2015